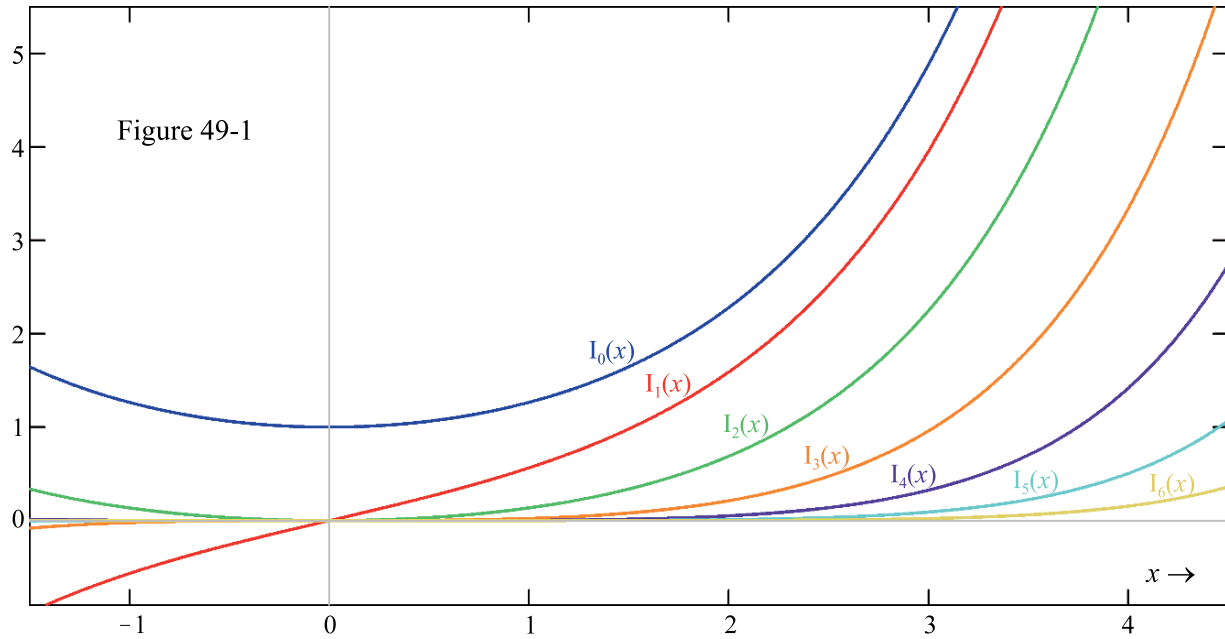




the solutions given in 50:3:4–6 do not provide a *complete* solution to the differential equations reported there. For a complete solution when $\nu = n$, each $I_{-\nu}$ term in those solutions is to be replaced by the corresponding K_n , as in equations 51:3:7–9.

Modified Bessel functions of integer order may be defined through a number of generating functions, including

$$49:3:1 \quad \exp\left(x \frac{1+t^2}{2t}\right) = \sum_{j=-\infty}^{\infty} I_j(x) t^j = I_0(x) + \sum_{j=1}^{\infty} I_j(x) [t^j + t^{-j}] = I_0(x) + I_1(x) \left[\frac{1+t^2}{t}\right] + I_2(x) \left[\frac{1+t^4}{t^2}\right] + \dots$$

and

$$49:3:2 \quad \exp(x \cos(\theta)) = \sum_{j=-\infty}^{\infty} I_j(x) \cos(j\theta) = I_0(x) + 2I_1(x) \cos(\theta) + 2I_2(x) \cos(2\theta) + \dots$$

The integral definitions

$$49:3:3 \quad I_n(x) = \frac{1}{\pi} \int_0^{\pi} \cos(n\theta) \exp(x \cos(\theta)) d\theta = \frac{x^n}{(2n-1)!!\pi} \int_0^{\pi} \sin^{2n}(\theta) \exp(x \cos(\theta)) d\theta$$

apply for all integer orders and, in addition, there are several, such as

$$49:3:4 \quad I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} \exp(\pm x \sin(\theta)) d\theta = \frac{1}{\pi} \int_0^{\pi} \cosh(x \sin(\theta)) d\theta = \frac{1}{\pi} \int_0^{\pi} \cosh(x \cos(\theta)) d\theta$$

and

$$49:3:5 \quad I_1(x) = \frac{1}{\pi x} \int_0^{2x} \frac{(x-t) \exp(x-t)}{\sqrt{2xt-t^2}} dt$$

that apply solely to the $n = 0$ or 1 cases. Definite integrals involving an angular integration variable, such as those in 49:3:4, may be adapted, as in

$$49:3:6 \quad I_0(x) = \frac{2}{\pi} \int_0^1 \frac{\cosh(xt)}{\sqrt{1-t^2}} dt$$

and

$$49:3:7 \quad I_1(x) = \frac{2x}{\pi} \int_0^1 \sqrt{1-t^2} \cosh(xt) dt$$

to a hyperbolic integrand.

Any modified Bessel function of positive integer order may be generated from its predecessor by a definite integration such as

$$49:3:8 \quad I_1(x) = x \int_0^1 t I_0(xt) dt, \quad I_2(x) = x \int_0^1 t^2 I_1(xt) dt, \quad I_3(x) = x \int_0^1 t^3 I_2(xt) dt, \quad \text{etc.}$$

Modified Bessel functions of orders 0 and 1 are generated by the operations of semiintegration or semidifferentiation [Section 12:14] applied to functions involving exponentials [Chapter 26]

$$49:3:9 \quad I_0(x) = \frac{\exp(-x)}{\sqrt{\pi}} \frac{d^{-1/2}}{dx^{-1/2}} \frac{\exp(2x)}{\sqrt{x}}$$

$$49:3:10 \quad I_1(x) + I_0(x) = \frac{\exp(-x)}{\sqrt{\pi}} \frac{d^{1/2}}{dx^{1/2}} \frac{\exp(2x)}{\sqrt{x}}$$

hyperbolic functions [Chapter 28]

$$49:3:11 \quad I_0(\sqrt{x}) = \frac{2}{\sqrt{\pi}} \frac{d^{1/2}}{dx^{1/2}} \sinh(\sqrt{x}) = \frac{1}{\sqrt{\pi}} \frac{d^{-1/2}}{dx^{-1/2}} \frac{\cosh(\sqrt{x})}{\sqrt{x}}$$

$$49:3:12 \quad I_1(\sqrt{x}) = \frac{1}{\sqrt{\pi x}} \frac{d^{-1/2}}{dx^{-1/2}} \sinh(\sqrt{x}) = 2\sqrt{\frac{x}{\pi}} \frac{d^{1/2}}{dx^{1/2}} \frac{\cosh(\sqrt{x})}{\sqrt{x}}$$

or error functions [Chapter 40]

$$49:3:13 \quad I_0(x) = \frac{\exp(x)}{\sqrt{2}} \frac{d^{1/2}}{dx^{1/2}} \operatorname{erf}(\sqrt{2x})$$

$$49:3:14 \quad I_1(x) = \frac{\exp(x)}{x} \frac{d^{-1/2}}{dx^{-1/2}} \left\{ \frac{\operatorname{erf}(\sqrt{2x})}{\sqrt{8}} - \frac{\sqrt{x} \exp(-2x)}{\sqrt{\pi}} \right\}$$

Modified Bessel functions of positive integer orders may be synthesized [Section 43:14] from the zero-order instance by

$$49:3:15 \quad I_0(2\sqrt{x}) \xrightarrow[n+1]{1} \frac{n!}{x^{n/2}} I_n(2\sqrt{x})$$

49:4 SPECIAL CASES

There are none. Be aware that, despite their integer subscripts, the modified spherical Bessel functions $i_n(x)$ [Section 28:13] do *not* correspond to modified Bessel functions of integer order, but to those having an order equal to half an odd integer.

49:5 INTRARELATIONSHIPS

The argument-reflection formula

$$49:5:1 \quad I_n(-x) = (-1)^n I_n(x) \quad n = 0, \pm 1, \pm 2, \dots$$

shows that the modified Bessel functions are even or odd according to the parity of their integer orders, as is also evident in Figure 49-1. The order-reflection formula

$$49:5:2 \quad I_{-n}(x) = I_n(x) \quad n = 0, \pm 1, \pm 2, \dots$$

asserts that modified Bessel functions of negative order merely duplicate their positive-ordered brethren, provided that the order is an integer.

Sufficient applications of the recursion formula

$$49:5:3 \quad I_{n+1}(x) = \frac{-2n}{x} I_n(x) + I_{n-1}(x) \quad n = 0, \pm 1, \pm 2, \dots$$

permit any modified Bessel function of integer order to be expressed in terms of $I_0(x)$ and $I_1(x)$. The general formula may be written

$$49:5:4 \quad I_n(x) = W_i^{(0)}(x) I_0(x) + W_i^{(1)}(x) I_1(x) \quad n = 2, 3, 4, \dots$$

where the W_i weighting functions are the polynomials are listed below that differ from the *Lommel polynomials*, discussed in Section 52:5 only by having different sign patterns.

	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$
$W_i^{(0)}(x)$	1	$\frac{-4}{x}$	$1 + \frac{24}{x^2}$	$\frac{-12}{x} - \frac{192}{x^3}$	$1 + \frac{144}{x^2} + \frac{1920}{x^4}$	$\frac{-24}{x} - \frac{1920}{x^3} - \frac{23040}{x^5}$
$W_i^{(1)}(x)$	$\frac{-2}{x}$	$1 + \frac{8}{x^2}$	$\frac{-8}{x} - \frac{48}{x^3}$	$1 + \frac{72}{x^2} + \frac{384}{x^4}$	$\frac{-18}{x} - \frac{768}{x^3} - \frac{3840}{x^5}$	$1 + \frac{288}{x^2} + \frac{9600}{x^4} + \frac{46080}{x^6}$

The right-hand side of the argument-multiplication formula

$$49:5:5 \quad I_n(bx) = b^n \sum_{j=0}^{\infty} \left(\frac{b^2 - 1}{2} \right)^j \frac{x^j}{j!} I_{j+n}(x) \quad n = 0, 1, 2, \dots$$

is an infinite series. The multiplier b is unrestricted but setting $b = \sqrt{3}$ or $\sqrt{5}$ gives series that are particularly simple. Other sums of modified Bessel functions of integer orders are

$$49:5:6 \quad \frac{1}{2} I_0(x) + I_2(x) + I_4(x) + I_6(x) + \dots = \frac{1}{2} \cosh(x)$$

$$49:5:7 \quad I_1(x) + I_3(x) + I_5(x) + \dots = \frac{1}{2} \sinh(x)$$

and

$$49:5:8 \quad \frac{I_2(x)}{2} + \frac{I_4(x)}{4} + \frac{I_6(x)}{6} + \dots = \frac{K_0(x) + \left\{ \gamma + \ln\left(\frac{1}{2}x\right) \right\} I_0(x)}{4}$$

and yet others arise by setting $t = 1$ or $t = -1$ in 49:3:1 or $\theta = \pi/2$ or π in 49:3:2. Euler's number γ [Chapter 1] occurs in formula 49:5:8, as does the zero-order case K_0 of the Macdonald function [Chapter 51].

49:6 EXPANSIONS

The power series, which may be written in terms of factorials or hypergeometrically,

$$49:6:1 \quad I_n(x) = \left(\frac{x}{2}\right)^n \sum_{j=0}^{\infty} \frac{(x^2/4)^j}{j!(j+n)!} = \frac{(\frac{1}{2}x)^n}{n!} \sum_{j=0}^{\infty} \frac{1}{(1)_j(n+1)_j} \left(\frac{x^2}{4}\right)^j \quad n=0,1,2,\dots$$

converges with alacrity. For $n = 0, 1, 2$, and 3 the first few terms in the series are

$I_0(x)$	$I_1(x)$	$I_2(x)$	$I_3(x)$
$1 + \frac{x^2}{4} + \frac{x^4}{64} + \frac{x^6}{2304} + \dots$	$\frac{x}{2} + \frac{x^3}{16} + \frac{x^5}{384} + \frac{x^7}{18432} + \dots$	$\frac{x^2}{8} + \frac{x^4}{96} + \frac{x^6}{3072} + \frac{x^8}{184320} + \dots$	$\frac{x^3}{48} + \frac{x^5}{768} + \frac{x^7}{30720} + \dots$

Asymptotic expansions for $I_n(x)$ follow by replacement of ν in 50:6:2 by n . Each can be written as a hypergeometric function [Section 18:14] or in terms of factorials. The $n = 0$ and $n = 1$ cases are

$$49:6:2 \quad I_0(x) \sim \frac{\exp(x)}{\sqrt{2\pi x}} \left[1 + \frac{1}{8x} + \frac{9}{128x^2} + \dots \right] = \frac{\exp(x)}{\sqrt{2\pi x}} \sum_{j=0}^{\infty} \frac{(\frac{1}{2})_j (\frac{1}{2})_j}{(1)_j} \left(\frac{1}{2x}\right)^j = \frac{\exp(x)}{\sqrt{2\pi x}} \sum_{j=0}^{\infty} \frac{[(2j)!]^2}{[j!]^3} \left(\frac{1}{32x}\right)^j$$

and

$$49:6:3 \quad I_1(x) \sim \frac{\exp(x)}{\sqrt{2\pi x}} \left[1 - \frac{3}{8x} - \frac{15}{128x^2} + \dots \right] = \frac{\exp(x)}{\sqrt{2\pi x}} \sum_{j=0}^{\infty} \frac{(\frac{-1}{2})_j (\frac{3}{2})_j}{(1)_j} \left(\frac{1}{2x}\right)^j = \frac{\exp(x)}{\sqrt{2\pi x}} \sum_{j=0}^{\infty} \frac{[(2j+1)!]^2}{j!(1-4j^2)} \left(\frac{1}{8x}\right)^j$$

being valid as $x \rightarrow \infty$.

49:7 PARTICULAR VALUES

All modified Bessel functions of integer order, except $I_0(x)$, take the value zero at $x = 0$ and have no other zeros. $I_0(x)$ equals unity at $x = 0$ and, when its argument is 2, equals the special number

$$49:7:1 \quad I_0(2) = \sum_{j=0}^{\infty} \left(\frac{1}{j!}\right)^2 = 2.2795\ 85302\ 33607$$

49:8 NUMERICAL VALUES

An elegant method of calculating values of modified Bessel functions of integer order is based on the property that, for a given argument x , the ratio R_j of the values of two modified Bessel functions of successive orders and identical arguments obeys the very simple recursion

$$49:8:1 \quad R_{j-1} = \frac{1}{R_j + (2j/x)} \quad \text{where} \quad R_j = \frac{I_{j+1}(x)}{I_j(x)}$$

This is a consequence of formula 49:5:3. Starting from the crude estimate $R_j \approx 2(J+1)/x$ of this ratio for a sufficiently large J , and employing 49:8:1 repeatedly, one may then develop the formula

$$49:8:2 \quad I_n(x) = \frac{R_{n-1}R_{n-2} \cdots R_1 R_0 \exp(x)}{\left(\left(\left(\left(\left(R_J + 1\right)R_{J-1} + 1\right)R_{J-2} + \cdots + 1\right)R_2 + 1\right)R_1 + 1\right)2R_0 + 1}$$

to calculate $I_n(x)$. This formula, which is an adaptation of *Miller's method* [Section 52:8], has its basis in the $t = 1$ version of equation 49:3:1.

The Miller method is not employed by *Equator*. The **modified Bessel function** routine (keyword **I**) described in Section 50:8, which caters to both noninteger and integer orders, is used. The I_v and I_n routines are identical.

49:9 LIMITS AND APPROXIMATIONS

All modified Bessel functions approach the same limit at large positive arguments

$$49:9:1 \quad I_n(x \rightarrow \infty) \rightarrow \frac{\exp(x)}{\sqrt{2\pi x}}$$

irrespective of their order, whereas they behave as power functions at small arguments

$$49:9:2 \quad I_n(x \rightarrow 0) \rightarrow \frac{1}{n!} \left(\frac{x}{2} \right)^n$$

49:10 OPERATIONS OF THE CALCULUS

Although all the formulas of Section 50:10 apply when $v = n$, the following differentiation and integration formulas are simpler than the general cases:

$$49:10:1 \quad \frac{d}{dx} I_0(x) = I_1(x)$$

$$49:10:2 \quad \frac{d}{dx} I_1(x) = I_0(x) - \frac{I_1(x)}{x}$$

$$49:10:3 \quad \int_0^x I_0(t) dt = \frac{\pi x}{2} [I_0(x) \mathbf{1}_{-1}(x) - I_1(x) \mathbf{1}_0(x)]$$

$$49:10:4 \quad \int_0^x I_1(t) dt = I_0(x) - 1$$

$$49:10:5 \quad \int_0^x t I_0(t) dt = x I_1(x)$$

$$49:10:6 \quad \int_0^x t I_1(t) dt = \frac{\pi x}{2} [I_1(x) \mathbf{1}_0(x) - I_0(x) \mathbf{1}_1(x)]$$

In these equations $\mathbf{1}_n$ represents a *modified Struve function* [Section 57:13].

An indefinite integral, specific to integer orders, is

$$49:10:7 \quad \int_0^x \exp(-t) I_n(t) dt = \exp(-x) \left[(n+x) I_0(x) + x I_1(x) + 2 \sum_{j=1}^{n-1} (n-j) I_j(x) \right] - n$$

Because the sum is empty if $n < 2$, the $n = 0$ and $n = 1$ cases of 49:10:7 are particularly simple.

The definite integrals listed in Section 50:10 apply when $v = n$, as do the differintegration formulas listed there. Special instances of the latter are the semioperations

$$49:10:8 \quad \frac{d^{1/2}}{dx^{1/2}} I_0(\sqrt{x}) = \frac{\cosh(\sqrt{x})}{\sqrt{\pi x}} \quad \text{and} \quad \frac{d^{-1/2}}{dx^{-1/2}} I_0(\sqrt{x}) = \frac{2}{\sqrt{\pi}} \sinh(\sqrt{x})$$

Among Laplace transforms of modified Bessel functions of integer order are:

$$49:10:9 \quad \int_0^\infty I_0(bt) \exp(-st) dt = \mathcal{L}\{I_0(bt)\} = \frac{1}{\sqrt{s^2 - b^2}}$$

$$49:10:10 \quad \int_0^\infty t^n I_n(bt) \exp(-st) dt = \mathcal{L}\{t^n I_n(bt)\} = \frac{(2n-1)!! b^n}{(s^2 - b^2)^{(2n+1)/2}}$$

49:11 COMPLEX ARGUMENT

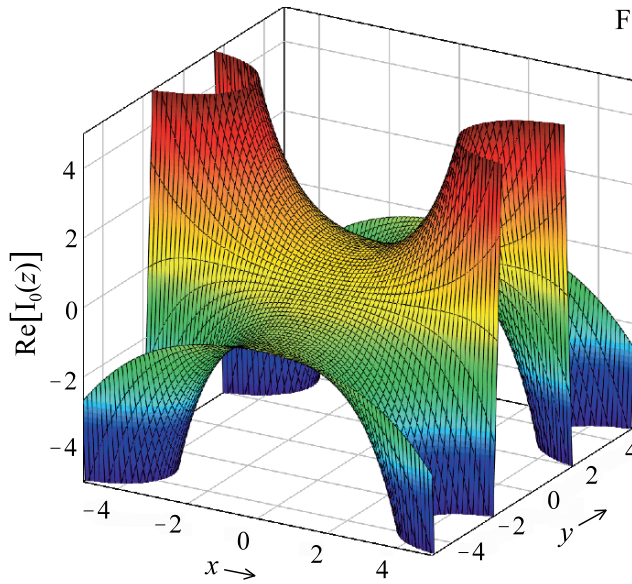


Figure 49-2

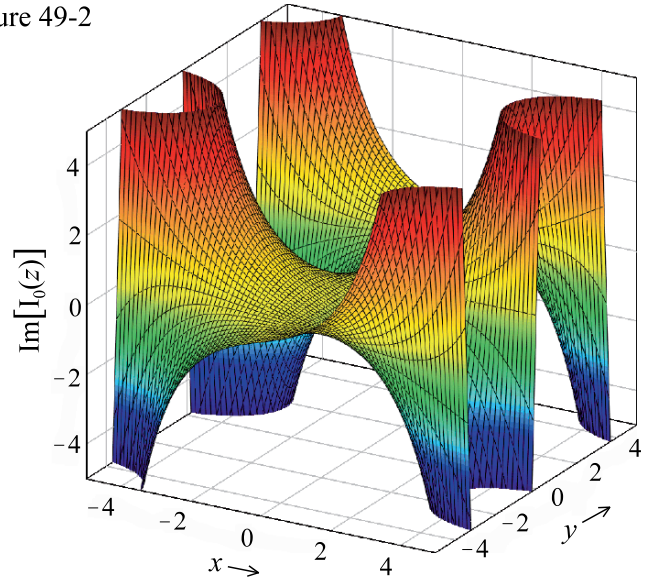


Figure 49-2 shows the real and imaginary parts of $I_0(z)$, where $z = x + iy$. Of course, taking sections through these three-dimensional graphs along the x -axis, where $y = 0$, gives the modified Bessel function of real argument. This implies

$$49:11:1 \quad \operatorname{Re}\{I_0(x + 0i)\} = I_0(x) \quad \text{and} \quad \operatorname{Im}\{I_0(x + 0i)\} = 0$$

It is instructive, however, also to cut along the y -axis, where $x = 0$, because the resulting profile is that of the (unmodified) Bessel function [Chapter 52]

$$49:11:2 \quad \operatorname{Re}\{I_0(0 + iy)\} = J_0(y) \quad \text{and} \quad \operatorname{Im}\{I_0(0 + iy)\} = 0$$

There is also interest in sectioning this figure diagonally, because this gives rise to Kelvin functions [Chapter 55]. Cutting along the $y = +x$ diagonal generates the following profiles

$$49:11:3 \quad \operatorname{Re}\{I_0(x + ix)\} = \operatorname{ber}_0(\sqrt{2}x) \quad \text{and} \quad \operatorname{Im}\{I_0(x + ix)\} = \operatorname{bei}_0(\sqrt{2}x)$$

whereas along the other diagonal, the imaginary component has the opposite sign:

$$49:11:4 \quad \operatorname{Re}\{I_0(x - ix)\} = \operatorname{ber}_0(\sqrt{2}x) \quad \text{and} \quad \operatorname{Im}\{I_0(x - ix)\} = -\operatorname{bei}_0(\sqrt{2}x)$$

The relationship between the various I , J , ber and bei cylinder functions, revealed for the zero order case in the equations above, may be further clarified by expressing the complex variable in polar $z = \rho \exp(i\theta)$, rather than in rectangular $z = x + iy$, coordinates. As illustrated diagrammatically in Figure 49-3, one then has

$$49:11:5 \quad I_0(z) = I_0(\rho \exp(i\theta)) = \begin{cases} I_0(\rho) & \theta = 0 \text{ or } \pi \\ \text{ber}_0(\rho) + i \text{bei}_0(\rho) & \theta = \frac{1}{4}\pi \text{ or } \frac{3}{4}\pi \\ J_0(\rho) & \theta = \pm \frac{1}{2}\pi \\ \text{ber}_0(\rho) - i \text{bei}_0(\rho) & \theta = \frac{3}{4}\pi \text{ or } \frac{1}{4}\pi \end{cases}$$

Figure 49-3

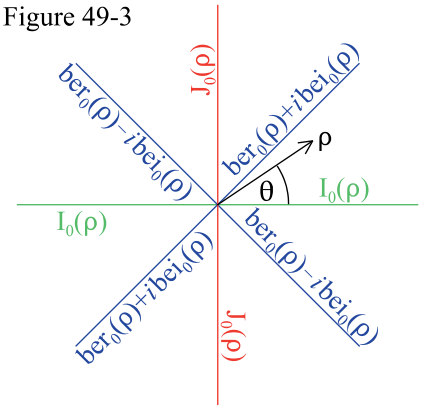
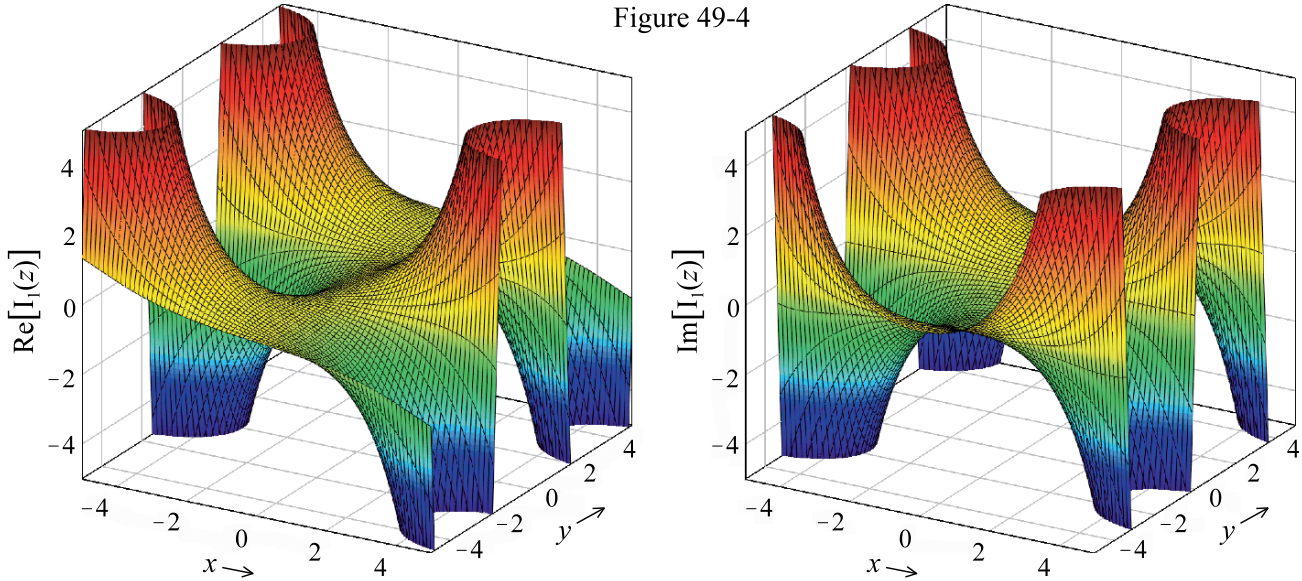


Figure 49-4 shows the real and imaginary parts of $I_1(x)$. Again, sectioning at right angles to the real axis gives rise to Bessel J functions, whereas scission at angles of 45° or 135° produces Kelvin ber_0 and bei_0 functions, though the relationships are somewhat different from the $n = 0$ case. The analogue of Figure 49-3 for the modified Bessel function of unity order is shown in the polar diagram Figure 49-5.

Figure 49-4



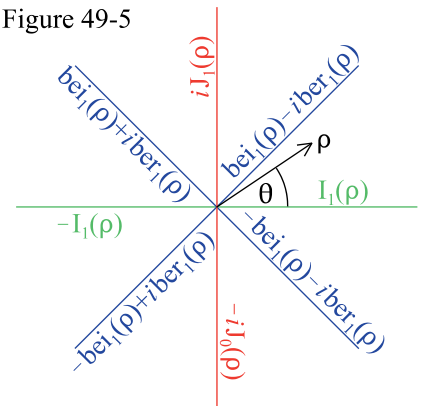
Properties of the general I_n function of complex argument may be deduced from equation 49:5:4 or from information in Section 50:11.

Interesting *window functions* [Section 9:13] arise from the inverse Laplace transformation of exponentially weighted modified Bessel functions of integer order. Two examples of these inverse transforms, which are nonzero for $0 < t < 2b$ but zero for $t > 2b$, are

$$49:11:6 \quad \int_{\alpha - i\infty}^{\alpha + i\infty} \exp(-bs) I_0(bs) \frac{\exp(ts)}{2\pi i} ds = \mathfrak{G}\{\exp(-bs) I_0(bs)\} \\ = \frac{u(t) - u(t - 2b)}{\pi \sqrt{t(2b - t)}}$$

and

Figure 49-5



$$49:11:7 \quad \int_{\alpha-j\infty}^{\alpha+j\infty} \exp(-bs) I_1(bs) \frac{\exp(ts)}{2\pi i} ds = \mathcal{G}\{\exp(-bs) I_1(bs)\} = \frac{[u(t) - u(t-2b)](b-t)}{\pi b \sqrt{t(2b-t)}}$$

49:12 GENERALIZATIONS

The next chapter addresses modified Bessel functions $I_\nu(x)$ of unrestricted order.

49:13 COGNATE FUNCTIONS

The modified Bessel functions of integer order are among the simplest of cylinder functions [Section 49:14], to all of which $I_n(x)$ is closely related.

The modified Bessel function of zero order and argument $2\sqrt{x}$

$$49:13:1 \quad I_0(2\sqrt{x}) = \sum_{j=0}^{\infty} \frac{1}{(1)_j (1)_j} x^j$$

is one of the *hypergeometric basis functions* [Section 43:14]. As such, it is the progenitor of a large number of named, and an infinite number of unnamed, functions. The functions listed in Table 18-5 and 18-6 are examples of the functions that may be synthesized from $I_0(2\sqrt{x})$ or its counterpart $J_0(2\sqrt{x})$.

49:14 RELATED TOPIC: cylinder functions generally

Functions that satisfy either the differential equation known as *Bessel's equation*

$$49:14:1 \quad \frac{d^2 f}{dx^2} + \frac{1}{x} \frac{df}{dx} - \left(\frac{v^2}{x^2} - 1 \right) f = 0$$

or that named *Bessel's modified equation*

$$49:14:2 \quad \frac{d^2 f}{dx^2} + \frac{1}{x} \frac{df}{dx} - \left(\frac{v^2}{x^2} + 1 \right) f = 0$$

or certain very similar equations, are known collectively as *cylinder functions*, because they arise in the solution of problems involving cylindrical (and other) geometries. The name *Bessel functions* is also sometimes applied generically to this collection of functions, though not in this *Atlas*. All cylinder functions are bivariate, with an order ν and an argument x . Interest focuses on positive values of ν and x , though the definitions of some cylinder functions remain valid for negative arguments or orders. Rational orders, such as $\nu = 1$, $\frac{3}{2}$, or $\frac{1}{3}$ are of especial interest because of their frequent occurrence in practical applications.

The relationship between the functions that satisfy 49:14:1 and those that satisfy 49:14:2 is exactly like that between, say, $\cos(x)$ and $\cosh(x)$ and it would therefore be appropriate to use the adjectives “circular” and “hyperbolic” to distinguish the two groups of functions. In practice the term “circular” is seldom employed, whereas “hyperbolic” is in frequent use to designate solutions of 49:14:2 and was so used in the first edition of this *Atlas*. Another name is *Bessel functions of imaginary argument*. Most usually, however, the adjective “modified” is employed, and this is the preferred descriptor in this second edition of the *Atlas* and for *Equator*. Thus a certain solution of equation 49:14:1 is known generally as the *Bessel function*; whereas the *modified Bessel function* or

hyperbolic Bessel function is the corresponding solution of 49:14:2.

Two solutions of the Bessel equation 49:14:1 are given the symbols $J_\nu(x)$ and $Y_\nu(x)$ and the names *Bessel function of the first kind* and *Bessel function of the second kind*. However, it is cumbersome to use such long names, so the unqualified name *Bessel function* is usually applied to the important $J_\nu(x)$ function, while $Y_\nu(x)$ is called the *Neumann function* (*Weber's function* is an alternative name and $N_\nu(x)$ is an alternative symbol). These two solutions have radically different properties near $x = 0$ and, on this basis, the Bessel function is described as “regular” and Neumann function as “irregular”. For the most part, regular functions adopt the value zero when their argument is zero, whereas irregular functions often approach $\pm\infty$ in the $x \rightarrow 0$ limit.

As discussed in Section 24:14, second order differential equations can have only two “linearly independent” solutions. However, linear combinations of such solutions are also solutions of the differential equation and, in solving Bessel’s equation, it is often convenient to use such combinations in preference to $J_\nu(x)$ and $Y_\nu(x)$. One such combination,

$$49:14:3 \quad \cos(\nu\pi)J_\nu(x) - \sin(\nu\pi)Y_\nu(x) = J_{-\nu}(x)$$

provides a solution which corresponds to a J function of negative order, but which is inapplicable (because $J_{-\nu}(x)$ then merely duplicates $J_\nu(x)$ or its negative) when ν is an integer. Two other combinations, useful in certain applications, are the *Bessel function of the third kind* or the *Hankel function*

$$49:14:4 \quad J_\nu(x) + iY_\nu(x) = H_\nu^{(1)}(x) \quad \text{and} \quad J_\nu(x) - iY_\nu(x) = H_\nu^{(2)}(x)$$

being known individually as *Hankel functions of the first and second kinds*.

The storyline is similar for the modified (or hyperbolic) family of functions. There is a regular solution $I_\nu(x)$, known as the *modified Bessel function of the first kind* or more simply as the *modified Bessel function*, as in earlier sections of this chapter. There is an irregular solution $K_\nu(x)$ described variously as the *modified Bessel function of the third kind*, *Bessel’s function of the second kind of imaginary argument*, the *modified Hankel function*, the *Basset function* and the *Macdonald function*, the last being the name used exclusively in this edition of the *Atlas*. In the modified family, the combination that yields functions of negative order is

$$49:14:5 \quad I_\nu(x) + \frac{2}{\pi} \sin(\nu\pi)K_\nu(x) = I_{-\nu}(x)$$

but is not the precise analogue of 49:14:3 because of the substantially different definitions of the Macdonald and Neumann functions.

In summary, the symbols and colloquial names adopted by this *Atlas* for the major cylinder functions are:

differential equation 49:14:1		differential equation 49:14:2	
regular solution	irregular solution	regular solution	irregular solution
$J_\nu(x)$	$Y_\nu(x)$	$I_\nu(x)$	$K_\nu(x)$
Bessel function	Neumann function	modified Bessel function	Macdonald function
[Chapters 52 and 53]	[Chapter 54]	[Chapters 49 and 50]	[Chapter 51]

Only when ν is half of an odd integer can solutions of equations 49:14:1 and 49:14:2 be expressed as functions simpler than cylinder functions. The names of such unusually simple functions incorporate the adjective “spherical” and use a lower case symbolism. Thus

$$49:14:6 \quad j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x) \quad [\text{see Section 32:13}]$$

$$49:14:7 \quad y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+\frac{1}{2}}(x) \quad [\text{see Section 32:13}]$$

$$49:14:8 \quad i_n(x) = \sqrt{\frac{\pi}{2x}} I_{n+\frac{1}{2}}(x) \quad [\text{see Section 28:13}]$$

and

$$49:14:9 \quad k_n(x) = \sqrt{\frac{\pi}{2x}} K_{n+\frac{1}{2}}(x) \quad [\text{see Section 26:13}]$$

and these *spherical cylinder functions* are addressed in earlier chapters. As well, many of these spherical cylinder functions represent special cases of the Struve function [Chapter 57] and its modified counterpart. There are also *spherical Hankel functions*, $h_n^{(1)}$ and $h_n^{(2)}$, defined similarly but not discussed in this *Atlas*.

Cylinder functions of order $\frac{1}{3}$ are termed *Airy functions* and are addressed in Chapter 56. Besides providing solutions of instances of equation 49:14:1, these functions also solve the simpler differential equation

$$49:14:10 \quad \frac{d^2 f}{dx^2} - x f = 0$$

known as *Airy's equation*.

Close cousins of the Bessel differential equations 49:14:1 and 49:14:2 are the two differential equations

$$49:14:11 \quad \frac{d^2 f}{dx^2} + \frac{1}{x} \frac{df}{dx} - \left(\frac{v^2}{x^2} \pm i \right) f = 0$$

which incorporate the imaginary i in place of unity. Solutions of these equations are known as *Kelvin functions* and are the subject of Chapter 55.

Not usually classified as cylinder functions, but closely related to them, are the *Struve functions* of Chapter 57. These participate in the solutions of inhomogeneous versions of equations 49:14:1 and 49:14:2; that is, analogues of these differential equations with nonzero right-hand sides.

CHAPTER 50

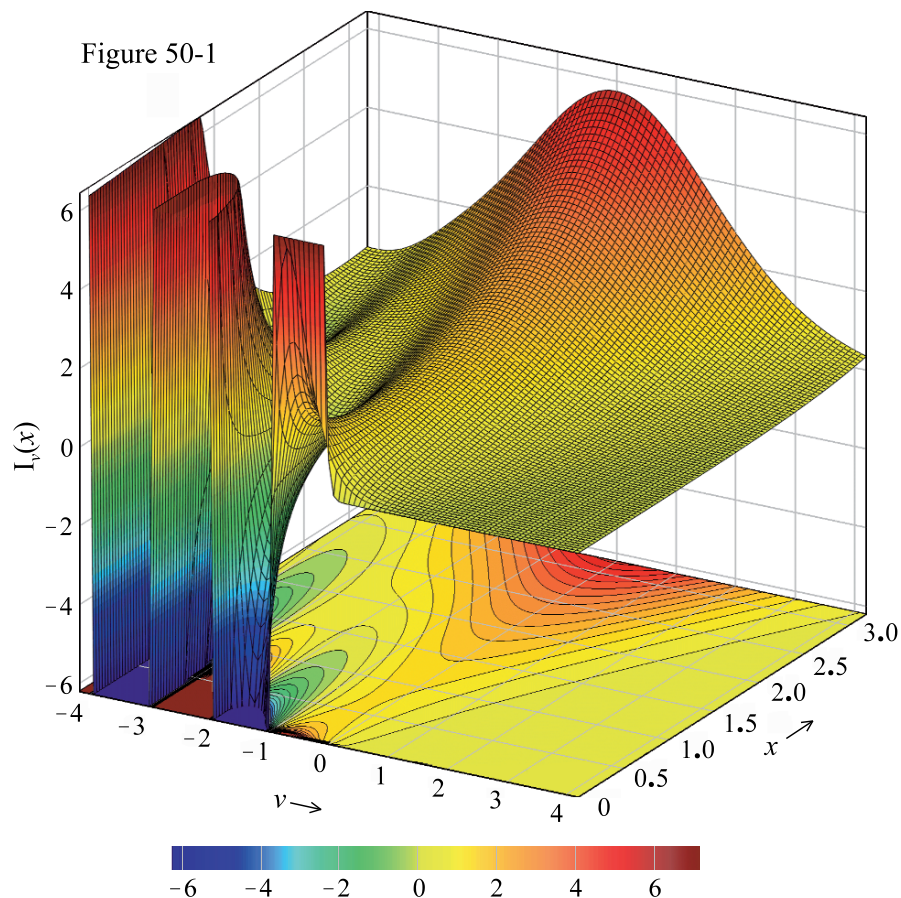
THE MODIFIED BESSEL FUNCTION $I_\nu(x)$ OF ARBITRARY ORDER

Because applications of the (unmodified) Bessel function predated and outnumber those of this chapter's function, the standard nomenclature treats the I function as subordinate to the J function, being designated as a "modified" version thereof. With equal, if not greater, validity one could regard the J function as a modification of the I function.

50:1 NOTATION

The names *hyperbolic Bessel function*, *Bessel function of imaginary argument*, and *Basset function* are alternative names. Sometimes the moniker *modified Bessel function of the first kind* is used.

The symbol $I_\nu(x)$ is standard, ν and x being termed the *order* and *argument* of the function. See Section 28:13 for the $i_\nu(x)$ symbol.



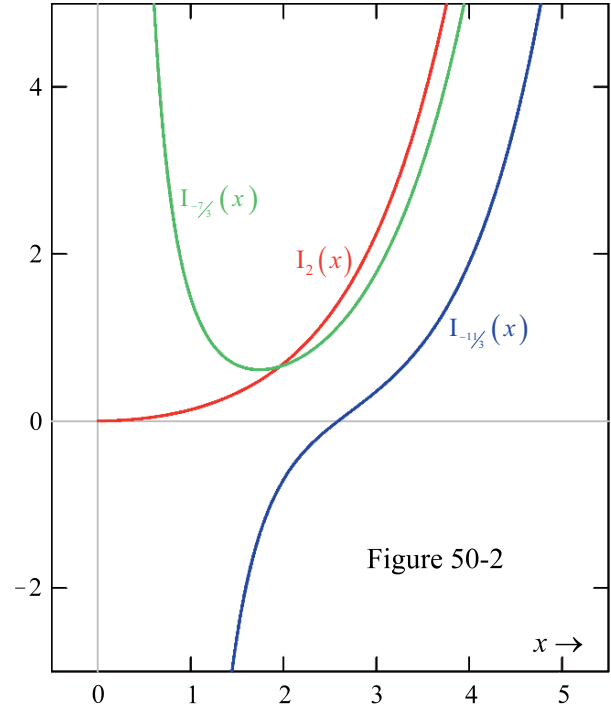
50:2 BEHAVIOR

Unless ν is an integer, the modified Bessel function $I_\nu(x)$ is complex when its argument is negative. Accordingly, x is treated as positive in most of this chapter and that is the only region depicted in Figures 50-1 and 50-2.

Irrespective of the order ν , the modified Bessel function rises steeply as x increases through more positive values. Closer to $x = 0$, the behavior of $I_\nu(x)$ falls into several categories as detailed in Section 50:7. One example from each of the three main categories is shown in Figure 50-2. For $\nu > 0$, such as the $\nu = 2$ case graphed, the function increases monotonically with x from $I_\nu(0) = 0$. For orders in the ranges $-1 < \nu < 0$, $-3 < \nu < -2$, $-5 < \nu < -4$, etc., such as the $\nu = -7/3$ case graphed, $I_\nu(0) = +\infty$, which implies that the modified Bessel function must display a minimum when viewed as a function of x ; such minimal values are positive and often occur not far from $x = 1$. For $-2 < \nu < -1$, $-4 < \nu < -3$, $-6 < \nu < -5$, etc., such as the $\nu = -11/3$ case graphed, $I_\nu(0) = -\infty$, and the modified Bessel function is once again a monotonic function of x , stretching in value from $-\infty$ to ∞ , with a single zero but without an extremum.

Unless ν is an integer, $I_\nu(x)$ is complex for negative x , whereas this is not the case for the composite function $I_\nu(x)/x^\nu$. This function equals $2^{-\nu}/\Gamma(1+\nu)$ at $x = 0$, and remains real for all real arguments. Being even, its properties for negative x are available through the reflection formula

$$50:2:1 \quad \frac{I_\nu(-x)}{(-x)^\nu} = \frac{I_\nu(x)}{x^\nu}$$



50:3 DEFINITIONS

The following integrals define the modified Bessel function, though only for $\nu > -1/2$:

$$50:3:1 \quad \frac{\sqrt{\pi} \Gamma(\nu + 1/2)}{(x/2)^\nu} I_\nu(x) = \int_{-1}^1 (1-t^2)^{(2\nu-1)/2} \exp(\pm xt) dt = \int_{-1}^1 (1-t^2)^{(2\nu-1)/2} \cosh(xt) dt$$

$$50:3:2 \quad \frac{\sqrt{\pi} \Gamma(\nu + 1/2)}{(x/2)^\nu} I_\nu(x) = \int_0^\pi \sin^{2\nu}(\theta) \exp\{\pm x \cos(\theta)\} d\theta = \int_0^\pi \sin^{2\nu}(\theta) \cosh\{x \cos(\theta)\} d\theta$$

A Kummer function [Chapter 47] reduces to a modified Bessel function whenever the former function's parameters are in a one-to-two ratio:

$$50:3:3 \quad \frac{(x/2)^\nu \exp(-x)}{\Gamma(1+\nu)} M\left(\frac{1}{2}+\nu, 1+2\nu, 2x\right) = I_\nu(x)$$

As well, a modified Bessel function may be defined through its complex algebraic relationship to an (unmodified) Bessel function as explained in Section 50:11, or via its series expansion 50:6:1.

Below are listed three differential equations and their solutions:

$$50:3:4 \quad x^2 \frac{d^2 f}{dx^2} + x \frac{df}{dx} - (x^2 + \nu^2) f = 0 \quad f = w_1 I_\nu(x) + w_2 I_{-\nu}(x)$$

$$50:3:5 \quad x \frac{d^2 f}{dx^2} + (2\nu + 1) \frac{df}{dx} - x f = 0 \quad f = \frac{w_1}{x^\nu} I_\nu(x) + \frac{w_2}{x^\nu} I_{-\nu}(x)$$

$$50:3:6 \quad x \frac{d^2 f}{dx^2} + (\nu + 1) \frac{df}{dx} - f = 0 \quad f = \frac{w_1}{(2\sqrt{x})^\nu} I_\nu(2\sqrt{x}) + \frac{w_2}{(2\sqrt{x})^\nu} I_{-\nu}(2\sqrt{x})$$

These differential equations, and several others, serve to define the modified Bessel function, but the listed solutions are incomplete when ν is an integer. See Section 49:3 for those cases.

The synthetic operations discussed in Section 43:14 can generate any modified Bessel function from the zero-order prototype:

$$50:3:7 \quad I_0(2\sqrt{x}) \xrightarrow{\nu+1} \frac{\Gamma(1+\nu)}{(x/2)^\nu} I_\nu(2\sqrt{x})$$

50:4 SPECIAL CASES

Modified Bessel functions of integer order are the subject of Chapter 49.

When the order is an odd multiple of $\pm 1/2$, modified Bessel functions are termed *spherical* and they reduce to hyperbolic cosines or sines, as discussed in Section 28:13.

Modified Bessel functions of order $\pm 1/3$ are closely related to the Airy functions of Chapter 56. The relationship

$$50:4:1 \quad I_{\pm 1/3}(x) = \frac{1}{2} \sqrt{\frac{3}{\hat{x}}} \left[\text{Bi}(\hat{x}) \mp \sqrt{3} \text{Ai}(\hat{x}) \right] \quad \hat{x} = \left(\frac{3x}{2} \right)^{2/3}$$

is obeyed. When the order is $\pm 2/3$, the corresponding formula involves the derivatives of the Airy functions

$$50:4:2 \quad I_{\pm 2/3}(x) = \frac{\sqrt{3}}{2\hat{x}} \left[\frac{d}{d\hat{x}} \text{Bi}(\hat{x}) \mp \sqrt{3} \frac{d}{d\hat{x}} \text{Ai}(\hat{x}) \right] \quad \hat{x} = \left(\frac{3x}{2} \right)^{2/3}$$

These formulas involve a radical change in argument.

A similar change in argument is sometimes helpful in expressing a modified Bessel function as a weighted sum of two or more parabolic cylinder functions [Chapter 46]. Such expressions exist whenever the order is a positive or negative odd multiple of $1/4$. The simplest cases are

$$\left. \begin{aligned} 50:4:3 \quad I_{\pm 1/4}(x) &= \frac{D_{-1/2}(-x') \mp D_{-1/2}(x')}{\sqrt{\pi x'}} \\ 50:4:4 \quad I_{\pm 3/4}(x) &= \frac{\mp 2D_{1/2}(x') + 2D_{1/2}(-x') \pm x'D_{-1/2}(x') + x'D_{-1/2}(-x')}{\sqrt{\pi(x')^3}} \end{aligned} \right\} \quad x' = 2\sqrt{x}$$

and others may be constructed via recursion 50:5:2.

50:5 INTRARELATIONSHIPS

A modified Bessel function of one order may be converted to another order by the synthetic process

$$50:5:1 \quad \frac{\Gamma(\nu+1)}{(x/2)^\nu} I_\nu(2\sqrt{x}) \xrightarrow{\nu'} \frac{\Gamma(\nu'+1)}{(x/2)^{\nu'}} I_{\nu'}(2\sqrt{x})$$

as prescribed in Section 43:14. As well, the recursion formula

$$50:5:2 \quad I_{\nu+1}(x) = I_{\nu-1}(x) - \frac{2\nu}{x} I_\nu(x)$$

may be applied to change the order of a modified Bessel function by an integer increment or decrement.

The two functions $I_\nu(x)$ and $I_{-\nu}(x)$ are identical if ν is an integer; otherwise

$$50:5:3 \quad I_{-\nu}(x) = I_\nu(x) + \frac{2\sin(\nu\pi)}{\pi} K_\nu(x)$$

where $K_\nu(x)$ is the Macdonald function of Chapter 51. This equation may be regarded as an order-reflection formula. There are similar order-reflection formulas, namely

$$50:5:4 \quad I_{\nu-\frac{1}{2}}(x) I_{-\nu-\frac{1}{2}}(x) = I_{\frac{1}{2}-\nu}(x) I_{\nu+\frac{1}{2}}(x) + \frac{2\cos(\nu\pi)}{\pi x}$$

and

$$50:5:5 \quad I_{-\nu}(x) I_{\nu\pm 1}(x) = I_\nu(x) I_{-\nu\mp 1}(x) + \frac{2\sin(\nu\pi)}{\pi x}$$

for the product of two modified Bessel functions whose orders sum to ± 1 .

On setting $b = i = \sqrt{-1}$, the argument-multiplication formula

$$50:5:6 \quad \frac{I_\nu(bx)}{b^\nu} = \sum_{j=0}^{\infty} \frac{I_{j+\nu}(x)}{j!} \left(\frac{(b^2-1)x}{2} \right)^j$$

generates the summation

$$50:5:7 \quad I_\nu(x) - \frac{x}{1!} I_{\nu+1}(x) + \frac{x^2}{2!} I_{\nu+2}(x) - \dots = \sum_{j=0}^{\infty} \frac{(-x)^j}{j!} I_{j+\nu}(x) = \frac{I_\nu(ix)}{i^\nu} = J_\nu(x)$$

whereas setting $b = \sqrt{3}$ leads to a similar series, but lacking the alternating signs.

$$50:5:8 \quad I_\nu(x) + \frac{x}{1!} I_{\nu+1}(x) + \frac{x^2}{2!} I_{\nu+2}(x) + \dots = \sum_{j=0}^{\infty} \frac{x^j}{j!} I_{j+\nu}(x) = \frac{I_\nu(\sqrt{3}x)}{3^{\nu/2}}$$

Replacing b in 50:5:6 by $1+(y/x)$ converts that equation into an argument-addition formula, though not a very useful one.

50:6 EXPANSIONS

The convergent power series expansion for the modified Bessel function

$$50:6:1 \quad I_\nu(x) = \frac{(x/2)^\nu}{\Gamma(1+\nu)} + \frac{(x/2)^{2+\nu}}{1!\Gamma(2+\nu)} + \frac{(x/2)^{4+\nu}}{2!\Gamma(3+\nu)} + \dots = \sum_{j=0}^{\infty} \frac{(x/2)^{2j+\nu}}{j!\Gamma(j+1+\nu)} = \frac{(x/2)^\nu}{\Gamma(1+\nu)} \sum_{j=0}^{\infty} \frac{1}{(1+\nu)_j (1)_j} \left(\frac{x^2}{4} \right)^j$$

has a hypergeometric formulation. There is also the asymptotic series, valid for large x

$$50:6:2 \quad I_\nu(x) \sim \frac{\exp(x)}{\sqrt{2\pi x}} \left[1 + \frac{\frac{1}{4}-\nu^2}{2x} + \frac{\left(\frac{9}{4}-\nu^2\right)\left(\frac{1}{4}-\nu^2\right)}{8x^2} + \frac{\left(\frac{25}{4}-\nu^2\right)\left(\frac{9}{4}-\nu^2\right)\left(\frac{1}{4}-\nu^2\right)}{48x^3} + \dots \right] = \frac{\exp(x)}{\sqrt{2\pi x}} \sum_{j=0}^{\infty} \frac{\left(\frac{1}{2}-\nu\right)_j \left(\frac{1}{2}+\nu\right)_j}{j!(2x)^j}$$

This series terminates when $\nu = \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \dots$ but the expansion is not exact even then. See Section 54:14 for more information about this series and its relation to the *auxiliary cylinder functions* $fc_\nu(x)$ and $gc_\nu(x)$.

Equation 50:5:7 shows how a J Bessel function may be expressed as a series of I's; the converse is also possible:

$$50:6:3 \quad I_\nu(x) = J_\nu(x) + \frac{x}{1!} J_{\nu+1} + \frac{x^2}{2!} J_{\nu+2} + \dots = \sum_{j=0}^{\infty} \frac{x^j}{j!} J_{j+\nu}(x)$$

50:7 PARTICULAR VALUES

The panel below, in which $m = 1, 2, 3, \dots$, shows the varied values adopted by $I_\nu(x)$ at $x = 0$:

	$\nu > 0$	$\nu = 0$	$-2m+1 < \nu < -2m+2$	$\nu = \pm m$	$-2m < \nu < -2m+1$
$I_\nu(0) =$	0	1	$+\infty$	0	$-\infty$

When $-2 < \nu < -1$, the modified Bessel function encounters a zero somewhere in the range $0 < x < 1.4$, but we know of no simple formula expressing the location of such a zero, except for the zero of $I_{-3/2}(x)$, which occurs at the value, close to 1.1997, of x such that $\coth(x) = x$. Similar zeros occur whenever ν lies between any two consecutive negative integers, the more negative being even.

On the other hand, when ν lies between two consecutive nonpositive integers, the more negative of which is odd, the modified Bessel function encounters no zero but rather a minimum. The minimum of $I_{-1/2}(x)$ occurs where $\coth(x) = 2x$.

50:8 NUMERICAL VALUES

With keyword **I**, *Equator's* [modified Bessel function](#) routine relies exclusively on equation 50:6:1. Negative arguments are not accepted unless ν is an integer.

50:9 LIMITS AND APPROXIMATIONS

Irrespective of its order, the modified Bessel function approaches the same limit

$$50:9:1 \quad I_\nu(x) \rightarrow \frac{\exp(x)}{\sqrt{2\pi x}} \quad x \rightarrow \infty$$

for large positive argument. For small positive arguments, however, the order has a dominant influence on the function's magnitude, as the limiting operation

$$50:9:2 \quad I_\nu(x) \rightarrow \frac{(x/2)^\nu}{\Gamma(1+\nu)} \quad x \rightarrow 0$$

demonstrates.